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MINERA QV - Smart Contract Security Audit

Project	**MINERA QV**
Contracts	Minera (MNRQV), an upgradeable ERC-20 token, and MineraPresale, an upgradeable presale and vesting contract
Chain	Polygon mainnet (chain 137)
Token	proxy 0xdBeedf87a99809B3c536c767602Aa3299438A603, implementation 0x3950d126741EE58554387059D6b0601e892Eb041
Presale	proxy 0x39043199A90D2FcFf71678C96AD01b17f70a7e66, implementation 0x04d10e3b96952B0401463C4a14144490C38D0Ba4
Owner	Gnosis Safe v1.4.1 0x40689AA60824EaaCFBE59d1d57d169De209f6108, threshold 2 of 3, owner of both contracts and upgrade controller of the token
Review	Initial audit completed 2026-07-20, re-reviews of the remediated code completed 2026-08-21 and 2026-09-04
Evidence	Polygon block 93,157,640 (2026-09-03), token state re-confirmed at block 93,381,033 (2026-09-07)
Standards	SWC Registry, CWE; review phases following NIST SP 800-115

1. Summary

MINERA QV consists of two upgradeable contracts on Polygon. The Minera token is an ERC-20 token with burning and signed approvals and a fixed supply of 1,000,000,000 MNRQV. MineraPresale sells MINERA for USDT at a price set by the owner, forwards every payment to a treasury and releases the purchased tokens through a vesting schedule with a cliff. Both contracts sit behind UUPS proxies and are owned by the same 2-of-3 Gnosis Safe.

We audited the contracts as first deployed and then reviewed two rounds of fixes. On 7 August 2026 both contracts were upgraded, and on 26 August 2026 the token was upgraded a second time. This report describes the code that is live today, token implementation V3 and the presale implementation of 7 August, and records every finding of the engagement with its final status. The deployed bytecode of every implementation matches the reviewed source.

A great deal has been fixed since the first audit. The initial token let a single externally owned account upgrade the contract, mint, pause, blacklist and freeze holders through an anti-bot regime. The initial presale let the same account withdraw the tokens owed to buyers and contained accounting defects that could pay buyers twice, strand part of every allocation or break the caps. All of these are resolved. The token now has no mint, pause, blacklist or transfer limit, its supply is fixed at the published 1,000,000,000, and both contracts are owned by a multisig instead of a single key.

What remains is mainly a question of control. Both contracts can still be upgraded immediately by the Safe, there is no timelock, and the Safe has only ever been operated by the same two of its three keys, both linked to the project. Everything the token upgrade removed could therefore be restored in a single transaction (T-001, PS-003, Medium). On the presale, three Low findings and some cleanup remain, and the contract still has to be funded with the 80 MINERA it owes to buyers before vesting can start.

Findings by severity and final status:

Severity	Found	Resolved	Mitigated	Open	Accepted or superseded
Critical	0	0	0	0	0
High	6	4	2	0	0
Medium	5	5	0	0	0
Low	6	3	0	3	0
Informational	22	13	2	3	4
Optimization	9	6	2	1	0
Total	**48**	**31**	**6**	**7**	**4**

Severities are those assigned when a finding was raised. The two mitigated High findings, T-001 and PS-003, are rated Medium today because ownership moved from a single account to a threshold multisig. Of the 48 findings, 22 concern the token and 26 the presale.

The main remaining points, in order of importance:

1. Upgrade authority (T-001, PS-003, Medium). The Safe can replace the code of either contract at once. All six Safe transactions were signed by owners 1 and 2, and owner 3 has never signed.

Until upgrades are delayed by a timelock and the keys are shown to be independent, holders and buyers rely on the Safe's restraint rather than on the code.

2. Presale funding (operational). The presale holds 0 MINERA against 80 MINERA allocated to buyers. The token can no longer mint, so the Safe must transfer the tokens from its own balance before startTGE is called, or every claim will revert.
3. Presale input and integration (PS-007, N-04, N-03, Low). There is no decimals-aware minimum investment, and the changed buy function signature has not been published.
4. Leftover code and storage (N-08, PS-010, N-05, Informational). Both contracts carry storage and code that is no longer used. On the token, the public getters still report the old limits and blacklist flags.

2. Scope and Method

2.1 Scope

Contract	File	Lines	SHA-256
Minera	Minera.sol (V3)	162	ae24e24dec76f57ca7eee661a287ef3ed8ce73c6e033c3f3a7b3161637ef6943
MineraPresale	Presale.sol	892	fa878726350e635c2fb4ee3e30a466579df85d81438efe52d2ed0603abb95a4c

Both files are identical to the verified sources of the live implementations. The earlier versions reviewed during the engagement are listed in section 3. All versions were compiled with solc 0.8.27, the optimizer at 200 runs, viaIR and EVM version istanbul, against OpenZeppelin Contracts and Contracts Upgradeable v5.4.0. The bundled OpenZeppelin files are identical to the official release.

Out of scope: the USDT token, the Gnosis Safe contract beyond its configuration and signatures, off-chain systems and front-ends, and any economic or tokenomics analysis.

2.2 Approach

The audit followed the phases planning, discovery, attack and reporting. In every round we combined the following:

- Manual review of each contract line by line and then as a system, covering access control and admin powers, initialization, upgrade safety, accounting, signature replay, front-running, denial of service, reentrancy, arithmetic and events.
- Static analysis of the initial code with four tools. Every result was triaged by hand as confirmed, mitigated or false positive.
- Dynamic testing with unit tests and proofs of concept, invariant tests, symbolic execution and stateful fuzzing on two engines. The harnesses were rewritten for every new version.
- On-chain verification: bytecode reproduction of every implementation, comparison of storage layouts across upgrades, reconstruction of the full upgrade history from archive state, recovery of the signers of every Safe transaction and reconciliation of every presale purchase.

For the remediation rounds we verified the chain first and compared the project's description of its changes with our results afterwards. No transactions were sent. Where a test exercises a power of the owner, we call it a capability demonstration; the word exploit is reserved for paths open to an untrusted caller.

2.3 Tools

Tool	Version	Used for
Foundry (forge, cast)	1.5.1	build, unit tests, invariants, storage layout, on-chain reads
Slither	0.11.5	static analysis, upgradeability and ERC-20 conformance checks
Aderyn	0.6.8	static analysis
Semgrep with Decurity rules	1.159.0	pattern analysis
Mythril	0.24.8	symbolic execution
Halmos	0.3.3	symbolic property checks
Echidna	2.3.2	stateful fuzzing
Medusa	1.5.1	stateful fuzzing
Sourcify		independent verification of the V3 source

2.4 Severity and status

Severity	Meaning
Critical	Direct and unbounded loss or creation of value that an untrusted party can cause
High	Material loss of user funds or control over holders' tokens, including unmitigated owner powers of that kind
Medium	Loss or lockout that needs specific conditions, or a defect that breaks a core guarantee without immediate loss
Low	Limited impact, or an impact that needs an unusual condition or a privileged action
Informational	No direct security impact: observability, documentation, code quality
Optimization	Gas or readability improvement

Owner powers are rated by what they allow and by how the key is held: the same power is rated higher for a single account than for a threshold multisig.

Resolved means the defect is gone or the agreed close criteria are met. Mitigated means the risk is reduced but the close criteria are not fully met. Open means no effective change. Risk-accepted means the project keeps the behaviour on purpose. Superseded means the item was folded into a newer finding.

2.5 Evidence

The current state was read at block 93,157,640 on 3 September 2026 through a public Polygon RPC node, and historical state through an archive node. The token state was re-confirmed unchanged on 7 September 2026 at block 93,381,033. Earlier rounds read the chain on 16 July 2026 (around block 90,320,615) and in August 2026 (block 92,397,256). The initial audit used the source delivered by the project, the first re-review the verified bundles from the block explorer and the second re-review Sourcify.

3. Deployment Context

	Minera token	MineraPresale
Proxy deployed	2026-04-02 07:02:45 UTC, block 84,996,664	2026-04-02 07:15:59 UTC, block 84,997,061
Initialization	inside the proxy deployment transaction	inside the proxy deployment transaction
Initial implementation	0xfb25a21ce2470ccf2fefd77a3810f528d7a167f5	0x83c9f2292b6b96a4b04bc2c8c7e107eb9206da78
Second implementation	0xFc3f30A225a63EF55063eB9D4ea15B922B78c193 (V2), from 2026-08-07	0x04d10e3b96952B0401463C4a14144490C38D0Ba4, from 2026-08-07, current
Third implementation	0x3950d126741EE58554387059D6b0601e892Eb041 (V3), from 2026-08-26, current	none
Owner	the Safe, which is also the upgrade controller	the Safe; two-step ownership, nothing pending
Live state	total supply 1,000,000,000 MNRQV, all held by the Safe	sale closed since 2026-08-03, TGE not started, 0 MINERA held against 80 allocated

Both proxies were initialized in their own deployment transactions, so their initialization could not be front-run. The token has seen only four transfers in its life: the genesis mint of 3,500,000,000 MNRQV to the deployer, a burn of 2,500,000,000, and the transfer of the remaining 1,000,000,000 to the Safe. No MINERA has ever been sent to the presale.

Date (UTC)	Event
2026-04-02	Both proxies deployed and initialized; genesis mint to the deployer
2026-05-01	Public sale started, end date set to 2026-10-31
2026-05-05 to 2026-06-18	Three purchases of 4 USDT in total for 80 MINERA, all by project-linked addresses
2026-06-10	Deployer burns 2,500,000,000 MNRQV; the Safe is created
2026-06-16	Deployer transfers 1,000,000,000 MNRQV to the Safe
2026-07-20	Initial audit delivered
2026-07-21	Ownership of both contracts and the token upgrade role move from the single account to the Safe
2026-08-03	The Safe sets the sale end to 2026-08-03 00:00, which closes the sale retroactively
2026-08-07	Presale upgraded; token upgraded to V2
2026-08-21	First re-review delivered
2026-08-26	Token upgraded to V3
2026-09-04	Second re-review delivered

4. How the System Works

4.1 Minera token

V3 is built from the OpenZeppelin upgradeable ERC-20, burnable, pausable and permit modules with Ownable, UUPS and a reentrancy guard. The token is called MINERA, has the symbol MNRQV and 18 decimals. The whole supply was minted at deployment, and no function can mint more. The transfer hook (_update) passes every transfer straight to the ERC-20 base contract and performs no checks; it also bypasses the pausable module, so transfers cannot be paused. Holders can burn their own tokens and approve spending by signature (EIP-2612) under the signing domain MINERA.

Upgrades are authorized when the caller equals the upgrade controller. That role is set to the new owner on every ownership transfer and cleared on renouncement, and the owner can reassign it directly. The owner can also sweep native currency or tokens that were sent to the token contract itself.

The earlier versions also had an owner mint, pause, blacklist and whitelist, a trading switch, and an anti-bot regime with transaction and wallet limits, a cooldown, per-period transaction counts and a holding period after large transfers. Most token findings came from these functions, and V3 removed all of them. Their storage variables are still present (N-08).

4.2 MineraPresale

The presale is built from Ownable2Step, a reentrancy guard, Pausable and UUPS.

- Buying: presale(investment, minimumTokensOut) checks the sale window, that the sale is active and TGE has not started, and that the accounting is initialized. It computes the tokens as investment times 1e18 divided by the price, reverts if they fall below the buyer's minimum, and checks the per-buyer cap (a separate cap applies to registered lifetime members) and the total cap against running token totals. The USDT goes straight from the buyer to the treasury.
- Claiming: claimPublic pays the vested amount minus what was already claimed from the contract's MINERA balance. State is updated before the transfer, and the function is protected against reentrancy.
- Vesting: nothing is released before the start plus the cliff. At the cliff a fixed percentage unlocks, and the rest follows linearly over the duration, optionally in steps of the interval. At the end of the duration the full allocation is available. startTGE sets the start to the current time, removes the cliff, applies a chosen initial unlock and closes the sale.
- Administration: the owner sets the price, investment token, caps, dates, treasury and, before vesting starts, the vesting schedule; registers members; pauses; upgrades; and can withdraw tokens, for MINERA only the part above what is still owed to buyers.

Live configuration: USDT (PoS) 0xc2132D05D31c914a87C6611C10748AEb04B58e8F with 6 decimals as investment token, a price of 0.05 USDT per MINERA, a total cap of 500,000,000 MINERA, and vesting over 365 days in 30-day steps after a 100-day cliff with 4 percent at the cliff. USDT never stays in the contract; the only asset it holds is the MINERA it pays claims from.

5. Privileged Roles

Role	Contract	Powers	Constraints
Owner (Safe)	Token	upgrade as upgrade controller, reassign the upgrade controller, transfer or renounce ownership, sweep assets held at the token address	immediate, no timelock; single-step ownership transfer
Owner (Safe)	Presale	upgrade; set price, investment token, caps, dates, treasury and the vesting schedule before it starts; register members; start the sale and TGE; pause; withdraw tokens, MINERA only above the amount owed	immediate, no timelock; two-step ownership transfer
Safe signers	Both	any two of three keys approve all of the above	owners 1 and 2 signed all six Safe transactions; owner 3 never signed

If two Safe keys are compromised or act together, they can upgrade either contract to arbitrary code. That would allow minting, freezing holders or taking the presale's MINERA and the buyers' entitlements, with no delay. Without an upgrade, changes to the presale price, investment token or treasury also take effect immediately.

6. Claims Check

Claim (token)	Verdict	Explanation
Can Mint	No	No mint function; supply fixed at 1,000,000,000. An upgrade could add it back (T-001).
Can Burn	Yes, own tokens only	burn, and burnFrom with an allowance
Can Blacklist	No	Removed in V3; the stored flags are not read. An upgrade could add it back (T-001).
Can Lock	No	No pause, trading switch or transfer limit. An upgrade could add them back (T-001).
Can Set Fees (higher as 25%)	No	No transfer fee, reflection or rebasing
Is Upgradeable	Yes	UUPS proxy, upgraded by the 2-of-3 Safe without delay

Property (presale)	Verdict	Explanation
Is upgradeable	Yes	UUPS proxy, upgraded by the Safe without delay (PS-003)
Owner can take buyers' MINERA	No	Withdrawals of MINERA are limited to the excess over what is owed (PS-002)
Buyer slippage protection	Yes	minimumTokensOut on every purchase (PS-004)
Minimum investment	No	No floor; dust purchases are accepted (PS-007, N-04)
Owner can change terms during the sale	Yes	price, caps, investment token, dates and treasury, immediately
Pause	Yes	Owner can pause buying and claiming (PS-006)
Reentrancy protection	Yes	On buying, claiming and emergency withdrawal
Initialization protected	Yes	Implementation disables initializers; both initializers are spent

Positive observations: both contracts use OpenZeppelin v5.4.0 without changes to the library code; the implementations cannot be initialized directly; permit signatures are bound to a per-holder nonce and the chain; the presale uses two-step ownership transfer and updates state before every token transfer; and every admin action in the current code emits an event.

7. Findings

ID	Contract	Severity	Title	Status
T-001	Token	Medium (was High)	Upgrade authority without timelock or separated role	Mitigated
T-002	Token	High	Trading switch and blacklist allowed holders to be trapped	Resolved
N-01	Token	High	Undisclosed increase of the supply cap to 100,000,000,000	Resolved
N-02	Token	High	Owner could freeze holders through unbounded limit setters	Resolved
T-003	Token	Medium	Genesis mint exceeded the maximum supply	Resolved
T-018	Token	Medium	Zero-value transferFrom could lock a holder's transfers	Resolved
T-004	Token	Low	Ownership transfer did not move upgrade authority	Resolved
T-005	Token	Low	Holding period after a large transfer could lock a victim	Resolved
PS-002	Presale	High	Emergency withdrawal could drain the tokens owed to buyers	Resolved
PS-003	Presale	Medium (was High)	Owner is the sole upgrade and parameter authority	Mitigated
PS-001	Presale	Medium	Buying again reset the claimed amount and allowed double claims	Resolved
PS-005	Presale	Medium	Caps were recomputed at the current price	Resolved
PS-017	Presale	Medium	Stepwise vesting never released the full allocation	Resolved
PS-004	Presale	Low	No slippage protection against price changes	Resolved
PS-007, N-04	Presale	Low	No decimals-aware minimum investment	Open
N-03	Presale	Low	Buy function signature changed without a published ABI	Open

IDs starting with T and PS were raised in the initial audit, N-01 to N-07 in the first re-review and N-08 in the second. The 22 informational and 9 optimization findings are listed in sections 7.3 and 7.4.

7.1 Minera token

T-001 Upgrade authority without timelock or separated role

Severity: Medium, centralisation (High in the initial audit). Status: Mitigated. Location: Minera.sol lines 27, 106 and 140-148 (upgradeController, _authorizeUpgrade, setUpgradeController). Classification: SWC-112, CWE-269. The presale shares this risk as PS-003.

The token is a UUPS proxy, and _authorizeUpgrade only checks that the caller is the upgrade controller. In the initial audit the owner and the upgrade controller were one externally owned account with no timelock. Since 21 July 2026 both roles are held by the 2-of-3 Safe. There is still no timelock, guard or module, so an upgrade takes effect in the transaction that approves it. Owners 1 and 2 have signed all six Safe transactions and owner 3 never has. Owner 1 was funded by the former single owner and owner 2 was previously the presale treasury, so the keys are not shown to be independent.

Impact: Two Safe keys can replace the token code without notice, for example to restore the mint and freeze functions removed in V3 or to move balances. The storage of the old functions is still in place, so a restored version would work immediately with the old values.

Recommendation: Transfer ownership of the token to a timelock, for example an OpenZeppelin TimelockController with the Safe as proposer and a delay that gives holders time to react. In V3 one transferOwnership moves both roles. A timelock that holds only the upgrade controller role is not enough, because the owner can reassign that role at once through setUpgradeController. Show that the Safe keys are held independently by replacing owners 1 and 2 with unrelated keys and having owner 3 sign. If no further upgrades are planned, renouncing ownership now removes the upgrade path permanently.

Resolution: Partly addressed. Moving ownership to a threshold multisig removed the single point of failure, which lowered the rating to Medium. The timelock and independent signers are outstanding.

T-002 Trading switch and blacklist allowed holders to be trapped

Severity: High, centralisation. Status: Resolved in V3. Location: initial version, transfer hook with setTradingEnabled and setBlacklist. Classification: SWC-105, CWE-284, CWE-841.

The initial transfer hook checked the trading switch and the blacklist before the whitelist exemption. The owner could let buyers in, then switch trading off or blacklist individual holders, while minting and burning kept working.

Impact: Holders could be prevented from ever selling or moving their tokens, which is the usual honeypot pattern.

Recommendation: Make the trading switch one-way and restrict or remove blacklisting.

Resolution: V2 made the trading switch one-way, but other freeze controls remained (N-02). V3 removed the trading switch, the blacklist, the whitelist and pause. A simulation against the live proxy confirmed that the old blacklist flags and limits still in storage are ignored.

N-01 Undisclosed increase of the supply cap to 100,000,000,000

Severity: High. Status: Resolved in V3. Location: V2, MAX_SUPPLY, PRESALE_ALLOCATION and mint. Classification: CWE-840.

The V2 upgrade raised MAX_SUPPLY from 1,000,000,000 to 100,000,000,000 tokens and PRESALE_ALLOCATION from 500,000,000 to 50,000,000,000, although the remediation letter stated that the cap had been left unchanged. Together with the owner mint, this opened 99,000,000,000 tokens of mint headroom. The wallet limit checked by mint could be raised by the owner without bound.

Impact: Every holder could be diluted up to 99 times without a further upgrade.

Recommendation: Restore the published cap, or disclose and bound the new figure, and assert the cap at the end of initialize.

Resolution: V3 restores MAX_SUPPLY to 1,000,000,000, sets INITIAL_SUPPLY and PRESALE_ALLOCATION to 500,000,000 each, asserts the cap at the end of initialize and removes mint. Live total supply equals the cap.

N-02 Owner could freeze holders through unbounded limit setters

Severity: High, centralisation. Status: Resolved in V3. Location: V2, setMaxTransactionAmount, setMaxWalletAmount, setCooldownPeriod, setLargeTransferProtection, pause and setBlacklist. Classification: CWE-284.

In V2 trading could no longer be switched off, but several owner setters still froze holders: a transaction or wallet limit of zero, a cooldown set to the maximum value, or a holding period of the maximum value triggered by any transfer. Pause and blacklist also remained.

Impact: The honeypot capability of T-002 was still available through other functions.

Recommendation: Bound every setter, or remove the owner's ability to freeze holders.

Resolution: V3 took the stronger option and removed all of these functions together with the anti-bot regime.

T-003 Genesis mint exceeded the maximum supply

Severity: Medium. Status: Resolved in V3. Location: INITIAL_SUPPLY, PRESALE_ALLOCATION, MAX_SUPPLY and initialize. Classification: SWC-101, CWE-682.

In the initial code initialize minted 3,500,000,000 tokens while MAX_SUPPLY was 1,000,000,000, because the genesis mint did not check the cap that mint enforced. This is what happened on the live proxy: the deployer received 3,500,000,000 MNRQV at deployment and burned 2,500,000,000 on 10 June 2026, which brought the supply down to the cap. V2 hid the inconsistency by raising the cap (N-01).

Impact: The supply cap did not hold at genesis, and the published tokenomics were wrong until the excess was burned. While supply was above the cap, mint could not be used.

Recommendation: Make the three constants consistent and assert the cap at the end of initialize.

Resolution: In V3 the genesis mint of INITIAL_SUPPLY plus PRESALE_ALLOCATION equals MAX_SUPPLY exactly, initialize ends with an assertion that total supply does not exceed the cap, and no other function can mint.

T-018 Zero-value transferFrom could lock a holder's transfers

Severity: Medium. Status: Resolved. Location: initial version, transfer hook. Classification: SWC-113, CWE-400.

Anyone could call transferFrom on a victim with an amount of zero, which needs no allowance. The call moved nothing but still updated the victim's cooldown and per-period transaction count.

Impact: Repeated at the cost of gas only, it kept a holder from sending their own tokens.

Recommendation: Ignore zero-value transfers before any anti-bot state is touched.

Resolution: V2 returned early on zero amounts, and V3 removed the anti-bot regime entirely.

T-004 Ownership transfer did not move upgrade authority

Severity: Low. Status: Resolved in V3. Location: Minera.sol lines 150-161 (transferOwnership, renounceOwnership). Classification: SWC-105, CWE-269.

transferOwnership and renounceOwnership did not touch the upgrade controller. After a handover the previous key could still upgrade, and a renouncement would have left the upgrade path open while looking like a final step. On 21 July 2026 the operator had to set the upgrade controller separately, three minutes before transferring ownership to the Safe.

Impact: Ownership and upgrade authority could drift apart, and "ownership renounced" would have been misleading.

Recommendation: Move the upgrade controller with ownership and clear it on renouncement.

Resolution: V3 sets the upgrade controller to the new owner in transferOwnership and clears it in renounceOwnership, with events.

T-005 Holding period after a large transfer could lock a victim

Severity: Low. Status: Resolved. Location: initial version, transfer hook and setLargeTransferProtection. Classification: SWC-113, CWE-400, CWE-841.

A transfer above the large-transfer threshold started a holding period for the recipient. Anyone could send a large amount to a holder and so block that holder's outgoing transfers for the holding period.

Impact: Unexpected lockouts for ordinary recipients, and a costly griefing path.

Recommendation: Do not let third parties start a lock on the recipient.

Resolution: V2 moved the clock to the sending side, and V3 removed the mechanism.

7.2 MineraPresale

PS-002 Emergency withdrawal could drain the tokens owed to buyers

Severity: High, custody. Status: Resolved. Location: Presale.sol lines 458-478 (emergencyWithdrawTokens). Classification: CWE-269.

In the initial code the owner could withdraw any token in any amount, including the MINERA reserved for buyers, and no event was emitted.

Impact: Buyers who had already paid could be left unable to claim.

Recommendation: Allow MINERA withdrawals only above the outstanding obligation and emit an event.

Resolution: The current code limits MINERA withdrawals to the balance minus allocated and not yet claimed tokens and emits an event. The obligation is seeded at 80 MINERA allocated and 0 claimed, which matches every purchase to the wei. Other tokens can still be withdrawn in full, which is intended.

PS-003 Owner is the sole upgrade and parameter authority

Severity: Medium, centralisation (High in the initial audit). Status: Mitigated. Location: Presale.sol line 891 (_authorizeUpgrade) and all owner setters. Classification: CWE-269.

Upgrades and all parameter changes are authorized by the owner alone. The owner was a single account in the initial audit and is the same Safe as on the token today. There is no timelock: price, investment token, treasury and emergency withdrawals take effect immediately, and the signers are not shown to be independent (see T-001).

Impact: Two Safe keys can upgrade the presale to arbitrary code and take the MINERA reserve and the buyers' entitlements, or change the economic terms without notice.

Recommendation: Move ownership to a timelock controlled by the Safe and show independent signers. The two-step ownership transfer of the presale makes this a safe operation.

Resolution: Partly addressed, as for T-001.

PS-001 Buying again reset the claimed amount and allowed double claims

Severity: Medium (High once funded and with vesting overlapping the sale). Status: Resolved. Location: Presale.sol lines 370-411 (presale) and 588-610 (claimPublic). Classification: CWE-682.

Every purchase in the initial code set the buyer's claimed amount back to zero while the allocation kept growing. A buyer who had claimed and then bought again could claim the earlier tokens a second time, paid out of the tokens owed to others. The proof of concept, and all three fuzzing engines, reproduced the sequence buy, claim, buy, claim.

Impact: Direct loss for other buyers and eventually failing claims. It was latent because the presale held no MINERA and vesting had not started.

Recommendation: Keep the claimed amount cumulative.

Resolution: The resets are removed, and claimed amounts only ever increase. The property that claimed never exceeds allocated now holds on Foundry, Echidna and Medusa.

PS-005 Caps were recomputed at the current price

Severity: Medium. Status: Resolved. Location: Presale.sol lines 389-396. Classification: CWE-682.

The initial code derived the tokens already sold from the USDT raised at the current price. A price increase freed headroom above the real cap, and a price decrease could make the subtraction revert and block all purchases. The per-buyer caps had the same problem.

Impact: Caps could be exceeded or the sale blocked by a normal price change.

Recommendation: Enforce the caps against running token totals.

Resolution: The global and per-buyer caps now use the cumulative allocated tokens. The price never changed on the live contract, so the defect never took effect.

PS-017 Stepwise vesting never released the full allocation

Severity: Medium. Status: Resolved. Location: Presale.sol lines 665-693 (_vestedAmount). Classification: CWE-682.

The initial vesting math rounded the elapsed time down to whole intervals even at the end. With the configured 365-day duration and 30-day interval, 5 days were never counted, so about 1.3 percent of every allocation would have stayed in the contract.

Impact: Every buyer would have lost part of the allocation, recoverable only by the owner.

Recommendation: Release the full amount once the duration has passed.

Resolution: The function returns the full allocation once the elapsed time reaches the duration, before rounding. A unit test and a symbolic check confirm it.

PS-004 No slippage protection against price changes

Severity: Low. Status: Resolved. Location: Presale.sol lines 370-388. Classification: SWC-114, CWE-362.

The price is set by the owner and was applied at execution time, with no minimum on the tokens received. A price change mined before a purchase gave the buyer fewer tokens.

Impact: Buyers could receive fewer tokens than expected.

Recommendation: Let buyers set a minimum.

Resolution: presale now takes minimumTokensOut and reverts below it, and price changes emit events. This changed the function signature (N-03).

PS-007 and N-04 No decimals-aware minimum investment

Severity: Low. Status: Open. Location: Presale.sol lines 384-388 and 810-817. Classification: CWE-682.

The initial code had a fixed minimum of 1 USDT expressed with 6 decimals, while the owner can switch to an investment token with different decimals. Instead of making the minimum decimals-aware, the remediation deleted it (recorded as N-04). The only remaining check is that a purchase yields at least one base unit of MINERA.

Impact: Dust purchases are accepted and inflate the participant count, and the investment token can still be changed during the sale.

Recommendation: Derive the minimum from the investment token's decimals when the token is set, or fix the investment token for the life of the sale.

N-03 Buy function signature changed without a published ABI

Severity: Low. Status: Open. Location: Presale.sol lines 370-373. Classification: CWE-758.

presale(uint256), selector 0xe6ab1434, was replaced by presale(uint256,uint256), selector 0xe5af14fa, to add slippage protection. The new ABI has not been published.

Impact: Front-ends or scripts that still call the old function will fail once the sale is reopened. The sale is closed only through its end date, which the owner can move.

Recommendation: Publish the new ABI before any reopening. Do not add a legacy wrapper without slippage protection, as that would reopen PS-004.

7.3 Informational findings

ID	Title	Status	Note
T-006	Max-wallet limit blocked receiving for wallets above it	Resolved	Limit removed in V3
T-007	Implementation did not disable its initializers	Resolved	Constructor disables initializers since V2
T-008	Emergency withdrawal sweeps the full balance	Mitigated	Event added; still no amount parameter. Concerns only assets sent to the token contract
T-009	Anti-bot timing relied on the block timestamp	Resolved	Timing logic removed in V3
T-010	Missing events on admin functions	Resolved	Events added
T-011	Permit domain name differed from the token name	Resolved	Domain is MINERA
T-012	No storage gap in the token	Risk-accepted	OpenZeppelin v5 uses namespaced storage; upgrades stay append-only
T-013	Distribution depended on the presale staying whitelisted	Resolved	No transfer limits remain
T-014	Code quality and conventions	Resolved	Pragma pinned, constants named
N-07	Holding-period mapping recorded sends under a receipt name	Superseded	Folded into N-08
N-08	Leftover anti-bot storage, unused Pausable base, events and errors	Open	Not reachable, but getters still show the old limits and blacklist flags. Remove the Pausable base (no slot shift) and keep the variables as named placeholders
PS-006	Pause modifiers without a way to pause	Resolved	pause and unpause added
PS-008	Missing events on setters	Resolved	Events on all setters
PS-009	startTGE could reset a configured schedule	Resolved	Inputs validated; removal of the cliff at TGE is documented
PS-010	Dead and commented-out code	Open	Private-sale blocks and an unused helper remain
PS-011	Misspelled public identifiers	Risk-accepted	getnontMemberCap and updateTreasure kept for compatibility
PS-012	Key state was not readable	Resolved	Getters added
PS-013	Sale and vesting windows rely on the block timestamp	Risk-accepted	Windows of days; validator drift is irrelevant
PS-014	Floating compiler pragma	Resolved	Pinned to 0.8.27
PS-015	No storage gap in the presale	Mitigated	Upgrades documented as append-only
PS-016	Zero vesting duration unlocked everything at the cliff	Resolved	Zero duration rejected by every setter
N-05	Dead private-sale storage added back (slots 20 to 25)	Open	Six unused slots; delete in a later upgrade or document

7.4 Optimization

ID	Title	Status
T-015	Custom errors instead of revert strings in the token	Resolved
T-016	Cache repeated storage reads in the transfer hook	Resolved
T-017	Minor gas and readability improvements in the token	Resolved
PS-G01	Custom errors instead of revert strings in the presale	Resolved
PS-G02	Optimize the member registration loop	Mitigated
PS-G03	Prefix increments	Resolved
PS-G04	Cache repeated storage reads in presale	Mitigated
PS-G05	Mark initialize as external	Open
PS-G06	Named constants instead of literals	Resolved

8. Code Quality

Both contracts are short and readable, and they use the OpenZeppelin modules as intended. The token became much simpler in V3: what is left is standard library behaviour plus a small owner surface, at the cost of leftover storage, events and errors from the removed functions (N-08). The presale uses custom errors, named constants and events throughout, but still carries commented-out private-sale code, an unused helper and storage for that code (PS-010, N-05). The public identifiers getnontMemberCap and updateTreasure keep their original spelling (PS-011).

9. Tool Results

9.1 Static analysis

Static analysis ran on the initial code; later versions were checked by review and by the test suites.

Tool	Results	Outcome
Slither	163 results, 62 in the audited files	confirmed results feed T-008, T-009, T-010, T-014, PS-010, PS-013, PS-017 and PS-011; ERC-20 conformance passed
Aderyn	1 high and 10 low issue classes	centralisation (T-001, T-002, PS-002, PS-003), missing events, pragma and gas items confirmed
Semgrep	65 hits	gas and style items (T-015 to T-017, PS-G01 to PS-G03); two-step ownership missing on the token
Mythril	1 low on the token, none on the presale	timestamp use in the OpenZeppelin permit deadline check, standard

False positives, each checked by hand: Aderyn's "contract locks Ether" (the presale cannot receive Ether), the reentrancy guard not being the first modifier (the preceding modifiers make no external calls), and the empty _authorizeUpgrade body (the modifier is the check). Slither's equality checks in the vesting math are deliberate sentinels, and Semgrep's possible underflow in claimableOf is guarded. The divide-before-multiply result in the vesting math was first dismissed as intended rounding and later confirmed as the cause of PS-017.

9.2 Symbolic execution and fuzzing

Halmos proved for the token that transfers preserve both the balances of the two parties and the total supply and that burns keep supply within the cap, and for the presale that vested never exceeds the allocation and that the full allocation vests at the end of the duration. The proof that vesting increases with time did not finish within the solver budget in any round. Echidna and Medusa reproduced the double claim of PS-001 on the initial code and pass on the current code.

9.3 Tests

The final suites pass in full: Foundry 27 of 27 tests (14 for the token, 13 for the presale), Halmos 5 of 6 checks with the known timeout, and Echidna and Medusa 4 of 4 properties each. Echidna and Medusa report the token's genesis assertion as never reached, which is expected because it only runs at deployment; a Foundry test covers it.

10. Invariants

Property	Contract	Tools	Initial code	Current code
Supply never exceeds MAX_SUPPLY and never grows after genesis	Token	Foundry, Halmos, Echidna, Medusa	Violated at genesis (T-003)	Pass
Supply changes only through mints and burns, accounted exactly	Token	Echidna, Medusa	Pass	Pass
Transfers preserve balances and supply	Token	Halmos	Pass	Pass
No mint, pause or freeze function is reachable	Token	Foundry	n/a	Pass
Transfers ignore the old stored limits and blacklist	Token	Foundry, live simulation	n/a	Pass
Permit signatures cannot be replayed	Token	Foundry	Pass	Pass
Claimed never exceeds allocated	Presale	Foundry, Echidna, Medusa	Fail (PS-001)	Pass
Emergency withdrawal leaves the MINERA owed to buyers	Presale	Foundry	Violated (PS-002)	Pass
Caps hold after a price change	Presale	Foundry	Violated (PS-005)	Pass
Full allocation claimable at the end of vesting	Presale	Foundry, Halmos	Violated (PS-017)	Pass
Vested never exceeds the allocation	Presale	Halmos	Pass	Pass
Vested amount increases with time	Presale	Halmos	n/a	n/a
Pause blocks buying and claiming	Presale	Foundry	Not reachable (PS-006)	Pass

11. On-Chain Verification

11.1 Bytecode

Implementation	Runtime size	Result
Token, initial 0xfb25a21c...	13,399 bytes	match
Token, V2 0xFc3f30A2...	13,916 bytes	match
Token, V3 0x3950d126...	11,008 bytes	match; Sourcify exact match
Presale, initial 0x83c9f229...	9,208 bytes	match
Presale, current 0x04d10e3b...	9,570 bytes	match

Each runtime was rebuilt from the verified source and is identical to the deployed code once the metadata hash and the UUPS self-address are normalised. The initial audit reported a residual of 5 bytes on the first implementations; it came from naming OpenZeppelin v5.3.0 instead of v5.4.0 and disappeared with the correct version.

11.2 Storage and initialization

The token's storage layout is identical in all three versions (slots 0 to 12). The presale kept slots 0 to 19 and appended new variables at slots 20 to 28. Both proxies are at initializer revision 2, and both initializers revert when called again. The V3 upgrade carried no initialization call.

11.3 Upgrade history and Safe transactions

The implementation history of both proxies was rebuilt by searching the ERC-1967 implementation slot across archive state, which would also reveal upgrades without events. The token was upgraded exactly twice and the presale once, as listed in section 3. The Safe has executed six transactions:

Nonce	Date (UTC)	Action	Signers
0	2026-06-13 08:55	sends 1 POL to the former single owner	owners 1 and 2
1	2026-07-21 11:00	accepts ownership of the presale	owners 1 and 2
2	2026-08-03 13:23	moves the sale end to 2026-08-03 00:00	owners 1 and 2
3	2026-08-07 11:08	upgrades the presale	owners 1 and 2
4	2026-08-07 11:39	upgrades the token to V2	owners 1 and 2
5	2026-08-26 09:12	upgrades the token to V3	owners 1 and 2

For every transaction the rebuilt Safe hash matches the executed one and the signatures recover to the listed owners. The Safe has no modules and no guard, and its owners and threshold have never changed.

12. Disclaimer

This report covers the two contracts named above as deployed at the blocks stated. It is based on the verified source, the deployed bytecode and the chain state at those blocks. Both contracts are upgradeable, and any later upgrade replaces the code reviewed here. The review did not include an economic or tokenomics analysis, the Gnosis Safe contract itself, the USDT token or off-chain systems. It is not investment advice, and investors should do their own research.



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